higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CONTROL
NQF LEVEL 4

(6050414)

15 March 2019 (Y-Paper)
13:00–16:00

Non-programmable calculators may be used.

This question paper consists of 5 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 Information flow is the transmission of signals back and forth.

1.2 Signal transfer equipment consists of transducers and transmission lines.

1.3 DCS stands for data control systems as used in screen schematics.

1.4 Cascade control is used in single-loop systems.

1.5 Override control is sometimes called selective control.

(5 × 1) **[5]**

QUESTION 2

Choose a word or item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (2.1–2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	Seven data points in a row that make a shift upward or downward.	A	control loop
2.2	One of the values that can be found on screen schematics.	B	manual control
2.3	A variable that no one can control.	C	run
2.4	Process control operation performed by people.	D	outlier
2.5	A computing device where a user enters a program using specialized software.	E	PLC
		F	set point
		G	trend
		H	data retrieval
		I	disturbance
		J	automatic control

(5 × 1) **[5]**

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QUESTION 3

3.1 Define the following terms:

3.1.1 Attribute data.

3.1.2 Histogram.

3.1.3 Preventative action.

(3 × 2) (6)

3.2 Products are formed according to certain specifications to ensure high quality end products.

State FOUR factors needed to be considered when product quality parameters are set. (4)

3.3 Differentiate between an X-chart and an R-chart. (2 × 2) (4)

3.4 Corrective action is a process of reacting to product problems, customer complaints or other nonconformities and fixing them.

State, in the correct order, the steps taken in the corrective action process. (5)

3.5 What does the acronym CAPA stand for? (1)

3.6 Pressure is defined as force per unit area exerted by a fluid (gas or liquid) on any surface.

Name and explain the THREE types of pressure commonly measured in industries. (3 × 2) (6)

3.7 Convert 149 °F to the Rankine scale. (4)

[30]

QUESTION 4

4.1 What is the common function of SCADA and DCS? (2)

4.2 State the main difference between SCADA and DCS. (2 × 2) (4)

4.3 Field instruments and sensors are components of DCS.

Name and describe the function of each of the three common examples of sensors used in the chemical industry. (3 × 2) (6)

4.4 Name THREE purposes of DCS screen schematics. (3)

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4.5 Make a sketch of the following symbols:

4.5.1 Cyclone.

4.5.2 Positive displacement pump.

(2 × 2) (4)

4.6 State FOUR reasons for data retrieval. (4)

4.7 Define the term *normal deviations*. (2)

[25]

QUESTION 5

5.1 Problem-solving methodology is one of the ways of eradicating problems in plants.

Explain the steps of problem solving in the correct order. (6)

5.2 What is the purpose of a start-up procedure in a simple process? (2)

5.3 Name FIVE factors that can affect a proper, safe and efficient way to start up a processing plant. (5)

5.4 Explain the importance of alarms in process plants. (2)

5.5 Name FIVE examples of deviations that may occur in process plants. (5)

[20]

QUESTION 6

6.1 Name THREE types of control systems. (3)

6.2 What does the acronym PLC stand for? (1)

6.3 Name FOUR uses of PLCs in plants. (4)

6.4 PLC is an industrial computer that provides control in a number of industrial situations.

Name THREE such situations. (3)

6.5 Name FOUR examples of software and gadgets that are used in process control to execute computations and logical operations with the control signals. (4)

[15]

TOTAL: 100

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